Q16. Sleuth Kit listing, metadata and extraction

Original question: Investigate a forensic disk image using The Sleuth Kit. List the available files and directories, examine the metadata of a designated file, or extract its contents as appropriate. Record and interpret the results.

Aim

List image files/directories, inspect the designated file's metadata and extract its contents when appropriate.

Requirements

The Sleuth Kit, supplied image/segments, partition offset and device sector size established as in Q10; selected file identifier. References: [R8], [R9], [R16]-[R18].

Procedure

Step 1 - Establish the filesystem context

```
mkdir -p "$HOME/EHF_Exam/Q16"
cd "$HOME/EHF_Exam/Q16"
IMAGE='<DISK_IMAGE_PATH>'
sha256sum "$IMAGE" | tee image_before.sha256
mmls "$IMAGE" | tee partitions.txt
START='<PARTITION_START_SECTOR>'
SECTOR_BYTES='<DEVICE_SECTOR_SIZE_BYTES>'
fsstat -b "$SECTOR_BYTES" -o "$START" "$IMAGE" > filesystem.txt
```

Use the actual partition's start sector, not its slot number or byte offset. For a confirmed filesystem-only image set START=0. Use supported container/segment handling; do not mount or alter the evidence. Keep the same offset and sector size for every command.

Step 2 - List names and select the designated file

```
fls -b "$SECTOR_BYTES" -o "$START" -r -p "$IMAGE" > files.txt
sed -n '1,100p' files.txt
```

`-r` recursively lists directories; `-p` includes full paths. Common prefixes include r/r for a regular-file entry and d/d for a directory; an asterisk can mark a deleted entry. The metadata identifier appears before the colon. Copy the complete identifier: NTFS entries may include an attribute type and ID such as 42-128-1. fls name entries and actual content availability are separate facts; deleted-directory recursion can be limited.

Step 3 - Inspect the selected metadata structure

```
META='<FILE_METADATA_ID>'
istat -b "$SECTOR_BYTES" -o "$START" "$IMAGE" "$META" \
  > selected_metadata.txt
cat selected_metadata.txt
```

`istat` displays the selected metadata structure: available allocation flags, size, links, ownership/permissions, timestamps and data-unit references vary by filesystem. Record timezone and timestamp meaning. Use the examiner's designated file, or document why a different available file was chosen.

Step 4 - Extract and verify a derivative file

```
OUT="$PWD/selected_file.bin"
icat -b "$SECTOR_BYTES" -o "$START" "$IMAGE" "$META" > "$OUT"
STATUS=$?
printf 'icat exit status: %s\n' "$STATUS"
stat -c '%n : %s bytes' "$OUT"
file "$OUT"
sha256sum "$OUT" | tee selected_file.sha256
sha256sum -c image_before.sha256
```

`icat` writes file content to standard output; > saves it locally and will replace an existing output. Use a new filename. Check STATUS before presenting an export as successful: redirection can create an empty/partial file even if icat fails. For a deleted file, a separately documented attempt may use `icat -r` with the same offset/identifier into a new output; success depends on surviving metadata/content. NTFS data streams require the appropriate full identifier. Do not use -s unless slack is explicitly requested, because it changes the output scope.

Expected output - illustrative

```
r/r 42-128-1: Users/student/Documents/note.txt
Selected size from metadata: 120 bytes
Extracted logical content size: 120 bytes
icat exit status: 0
```

These identifiers and sizes are synthetic. Your identifier may be a simple inode instead. Do not paste 42-128-1 unless it is actually listed in your image.

Interpretation, observation and result

Record full source path, identifier/stream, partition offset, filesystem, allocation state, metadata times, extraction status, output size and hash. Compare logical size and content with metadata; a successful extraction alone cannot guarantee intact deleted content. Metadata block addresses and file identifiers are different quantities. The result is a traceable listing, metadata record and derivative file, or a documented reason extraction could not be completed.

Common errors

- Invalid inode/identifier: copy the complete fls metadata identifier from the correct volume.
- Cannot determine filesystem: verify START, SECTOR_BYTES and image format.
- Empty extraction: inspect stderr/status, streams and surviving allocation; do not claim recovery just because the output exists.

Viva questions

- What is the purpose of fls? Listing filesystem directory entries and metadata identifiers.
- What is istat used for? Inspecting one metadata structure and its available attributes.
- Why redirect icat output? It writes content to stdout; redirection saves the derivative file.
- Does a deleted entry guarantee recovery? No; its content may have been overwritten or trimmed.